



Q.P.

Sardar Patel Institute of Technology

Bhavan's Campus, Munshi Nagar, Andheri (West), Mumbai-400058, India
(Autonomous College Affiliated to University of Mumbai)

End Semester Examination

Nov – Dec 2017

Max. Marks: 100

Class: T.E.

Course Code: ITC502

Name of the Course: Operating Systems

Duration: 3Hrs

Semester: V

Branch: IT

Instruction:

- (1) All questions are compulsory.
- (2) Draw neat diagrams.
- (3) Assume suitable data if necessary.

Q No.		Max. Marks	CO
Q.1 (a)	What is the need of system call? Following code will create how many process and why? Show process created with neat diagram. <pre>int main() { int i; for(i= 0; i < 3; i++) fork(); return(0); }</pre>	5	CO1
Q.1 (b)	What are the 4 advantages and a disadvantage of layered architecture of OS ?	5	CO1
Q.1 (c)	What is RAID? Why it is needed? Explain RAID 10 with neat diagram along with its advantage and disadvantages.	10	CO2
Q.2 (a)	What resources are used when a thread is created? How do they differ from those used when a process is created? Differentiate between user level thread and kernel level thread for 5 different points. OR Explain process state transition diagram ,also write down in each state what will be the contents of PCB with respect to the state of the process ? Why process state diagram exists?	10	CO2

Q.2 (b)	Consider the following set of processes, with the length of the CPU burst time given in milliseconds: <table><tr><th>Process</th><th>Burst time</th><th>Priority</th></tr><tr><td>P1</td><td>10</td><td>3</td></tr><tr><td>P2</td><td>1</td><td>1</td></tr><tr><td>P3</td><td>2</td><td>3</td></tr><tr><td>P4</td><td>1</td><td>4</td></tr><tr><td>P5</td><td>5</td><td>2</td></tr></table> The processes are assumed to have arrived in the order P1,P2,P3,P4,P5, all at time 0. a. Draw Gantt chart illustrating the execution of these processes using FCFS,SJF, and nonpreemptive priority (a smaller priority number implies a higher priority) scheduling. b. What is the turnaround time of each process for each of the scheduling algorithms in part a? c. What is the waiting time of each process for each of the scheduling algorithms in part a? d. Which of the schedules in part a results in the minimal average waiting time (over all processes)?	Process	Burst time	Priority	P1	10	3	P2	1	1	P3	2	3	P4	1	4	P5	5	2	10	CO2																														
Process	Burst time	Priority																																																	
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Q.3 (a)	An operating system has 3 types of resources with the following quantities of each type: Resource Type: R1 R2 R3 Quantity of that type: 13 8 10. Show the calculations of the Banker's algorithm to determine whether or not this system is in a safe state. <table><tr><th rowspan="2">Thread</th><th colspan="3">Max</th><th colspan="3">Allocation</th></tr><tr><th>R1</th><th>R2</th><th>R3</th><th>R1</th><th>R2</th><th>R3</th></tr><tr><td>T0</td><td>2</td><td>8</td><td>0</td><td>1</td><td>1</td><td>0</td></tr><tr><td>T1</td><td>5</td><td>2</td><td>2</td><td>4</td><td>1</td><td>1</td></tr><tr><td>T2</td><td>4</td><td>3</td><td>9</td><td>1</td><td>1</td><td>7</td></tr><tr><td>T3</td><td>2</td><td>2</td><td>3</td><td>2</td><td>1</td><td>1</td></tr><tr><td>T4</td><td>2</td><td>1</td><td>3</td><td>2</td><td>0</td><td>0</td></tr></table> OR What is 'wait for' graph scheme? What is its drawback? Explain deadlock detection algorithm.	Thread	Max			Allocation			R1	R2	R3	R1	R2	R3	T0	2	8	0	1	1	0	T1	5	2	2	4	1	1	T2	4	3	9	1	1	7	T3	2	2	3	2	1	1	T4	2	1	3	2	0	0	10	CO3
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Q.3 (b)	Compare Monitor and Semaphore for 4 different points and write down their use.	5	CO3																																																
Q.3 (c)	What does the positive and negative value of counting semaphore indicates? A counting semaphore was initialized to 10. Then 6P (wait) operations and 4V (signal) operations were completed on this semaphore. The resulting value of the semaphore is—. (a) 0 (b) 8 (c) 10 (d) 12	5	CO3																																																

Q.4 (a)	1) Consider a system with 80% hit ratio, 50 nano-seconds time to search the associative registers , 750 nano seconds time to access memory. Find the time to access a page a. When the page number is in associative memory. b. When the time to access a page when not in associative memory. c. Find the effective memory access time. 2) What is the disadvantage of paging. ? Consider a logical address space of 8 pages of 1024 words mapped into memory of 32 frames. a. How many bits are there in the logical address ? b. How many bits are there in physical address ? OR Consider main memory (MM) of 64MB, logical address is 32 bit, page size is 4KB. Compute total space wasted in maintaining page table by the system.	10	CO4
Q.4 (b)	Given page reference string: 7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3. Take 3 page frames. Calculate page fault ratio for LRU and optimal page replacement policy.	10	CO4
Q.5 (a)	Cylinder requests are 55, 58, 39, 18, 90, 160, 150, 38, 184 . There are 200 cylinders numbered from 0 – 199. The disk head starts at number 100. Calculate total head movement and average seek length using SCAN and CSCAN disk scheduling algorithm. Assume head moves towards increasing track number.	10	CO2
Q.5 (b)	Explain contiguous file allocation method with example and neat diagram. Write down its advantages and disadvantages. OR b) Explain three methods of record blocking in file management with neat diagrams.	10	CO5